Lingheng Meng

Postdoctoral Fellow, CSIRO Technology

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Professional Summary

Trained at the University of Waterloo (PhD, 2023), I am currently a CSIRO Early Research Career (CERC) Postdoctoral Fellow in CSIRO Technology (Clayton) and an Adjunct Lecturer in Electrical and Computer Systems Engineering at Monash University. My central research question is how to make robots that learn from and act alongside people robust and trustworthy for real-world deployment. I pursue this through robot learning (primarily deep reinforcement learning and preference learning) paired with human-robot interaction (HRI), applied across platforms spanning manipulation, legged locomotion, and interactive architectural installations. This combines algorithmic and empirical research with real-world challenges such as partial observability and the sim-to-real gap, explored across simulation, in-lab HRI studies, and field deployment with real users.

- **Research impact:** 544 citations, h-index 8, i10-index 8 (Google Scholar, 16 October 2026); publications in *ACM Transactions on Human-Robot Interaction (THRI)*, *Neural Networks*, *Transportation Research Part C*, IROS, ICPR.
- **Teaching & supervision:** co-designing and teaching Monash University's ECE4078 Intelligent Robotics unit (∼200 students, Semester 2, 2026), including Reinforcement Learning and HRI sessions. Co-supervising 2 PhD students at Monash University (both with oral presentations, RO-MAN 2026 and ACM/IEEE HRI 2026) and supervised 15 Final Year Project (FYP, Australia's undergraduate equivalent of a U.S. capstone/senior design project) students across 5 projects.
- **Physical & simulation infrastructure:** I set up and maintain the hardware and compute that my research, students, and colleagues rely on, not just the algorithms, including the Franka Panda platform, a sim-to-real pipeline, and a GB300 (NVIDIA Grace Blackwell) AI workstation at CSIRO, and national high-performance computing (HPC) allocations supporting Prof. Kulić's Monash Robotics group.

Education

2017–2023 **PhD in Electrical and Computer Engineering**, *University of Waterloo*, Waterloo, Ontario, Canada
Thesis: *Learning to Engage: An Application of Deep Reinforcement Learning in Living Architecture Systems*. Advisers: Prof. Dana Kulić and A/Prof. Rob Gorbet.
Committee: A/Prof. Urs Hengartner (Chair); A/Prof. Edith Law; Prof. Stephen L. Smith, Canada Research Chair in Autonomous Systems; A/Prof. Mark Crowley; Matthew Gombolay, Associate Professor, Georgia Institute of Technology (External Examiner).
Focus: Reinforcement Learning, Preference Learning, HRI, Large-scale Interactive Systems.
Cumulative Marks: 85.50/100.

2022–2023 **Visiting PhD Student, Electrical and Computer Systems Engineering**, *Monash University*, Clayton, Victoria, Australia
Continued PhD thesis research. Adviser: Prof. Dana Kulić.

2014–2017 **MEng in Computer Application Technology**, *China University of Mining and Technology*, Xuzhou, Jiangsu, China
Thesis: *Research on Autoencoder and Its Application*. Adviser: Prof. Shifei Ding.
Cumulative Marks: 89.18/100.

2010–2014 **BEng in Computer Science and Technology**, *China University of Mining and Technology*, Xuzhou, Jiangsu, China
Thesis: *3D Scene Perception System Based on Single Static Image*.
Cumulative GPA: 3.79/4.0, Ranking 5/165 (Graduated as Outstanding Graduate).

Work Experience

Apr 2024–Now **CSIRO Early Research Career (CERC) Fellowship**, *CSIRO: Commonwealth Scientific and Industrial Research Organisation*, Clayton, Victoria, Australia
Conducting research in Human-Robot Interaction and Robot Learning, including the Patchable Preference Learning project and multi-view preference-based learning work. Co-supervisor for Monash University–CSIRO PhD students (oral presentations at RO-MAN 2026 and ACM/IEEE HRI 2026). Administer two national HPC allocations (Pawsey Setonix, NCI Gadi) held by Prof. Dana Kulić's Monash Robotics group, including onboarding 20+ researchers/students and negotiating quota top-ups. Led procurement and manage the lab's GB300 AI workstation, including adding a dedicated display GPU (RTX PRO 6000 Blackwell) for local, ray-traced rendering of HRI interfaces without network-transfer latency. Fellowship supervisor: Prof. Dana Kulić; line manager: Dr. Leimin Tian.

Aug 2024–Now **Adjunct Lecturer, Electrical and Computer Systems Engineering**, *Monash University*, Clayton, Victoria, Australia
Teaching Monash University's undergraduate ECE4078 Intelligent Robotics unit, co-supervising Final Year Project (FYP) and PhD students, milestone assessment panels, and research-training contributions in the Faculty of Engineering.

Aug 2023–Apr 2024 **Postdoctoral Research Fellow, Electrical and Computer Systems Engineering**, *Monash University*, Clayton, Victoria, Australia
Worked on upgrading robot learning infrastructure at Monash Robotics and began extending my PhD-era multi-step bootstrapping research toward the robustness work later published in *Neural Networks* (2026). Supervisors: Prof. Dana Kulić and Dr. Michael Burke.

May 2023–Jul 2023 **Visitor, Electrical and Computer Systems Engineering**, *Monash University*, Clayton, Victoria, Australia
Worked on Road Safety Innovation Fund project. Host: Prof. Dana Kulić.

Research Interests

Human-Robot Interaction; Deep Reinforcement Learning; Preference Learning; Learning-Based Control; Imitation Learning; Inverse Reinforcement Learning; Vision-Language-Action (VLA) Models for Robotics; Robot Learning for Assistive Robotics; Sim-to-Real Transfer; Interactive/Adaptive Robotic Systems; Affective and Personalized Human-Machine Interaction; Road Safety Vision; Developmental Robotics; Intrinsic Motivation in Robots.

Publications

Journals

2026 **L. Meng**, R. Gorbet, M. Burke, D. Kulić. Multi-step first: A lightweight deep reinforcement learning strategy for robust continuous control with partial observability. *Neural Networks*, 199, 108521.

2025 M. Li, B. Beck, T. Rathnayake, **L. Meng**, Z. Chen, A. Cosgun, X. Chang, D. Kulić. A benchmark for cycling close pass detection from video streams. *Transportation Research Part C: Emerging Technologies*, 174, 105112.

2021 **L. Meng**, D. Lin, A. Francey, R. Gorbet, P. Beesley, D. Kulić. Learning to Engage with Interactive Systems: A Field Study on Deep Reinforcement Learning in a Public Museum. *ACM Transactions on Human-Robot Interaction*, 10(1), 1–29.

2018 **L. Meng**, S. Ding, N. Zhang, J. Zhang. The Research of Stacked Denoising Sparse Autoencoder. *Neural Computing and Applications*.

2017 S. Ding, **L. Meng**, Y. Han, Y. Xue. A Review on Feature Binding Theory and Its Applications in Perceptual Learning. *Cognitive Computation*, 9(2), 194–206.

2017 **L. Meng**, S. Ding, Y. Xue. Research on denoising sparse autoencoder. *International Journal of Machine Learning and Cybernetics*, 8(5), 1719–1729.

2014 H. Jia, S. Ding, **L. Meng**, S. Fan. A density-adaptive affinity propagation clustering algorithm based on spectral dimension reduction. *Neural Computing & Applications*, 25(7-8), 1557–1567.

Conferences

2026 E. Fashae, M. Burke, L. Tian, **L. Meng**, P. Carreno-Medrano. Explaining Why Things Go Where They Go: Interpretable Constructs of Human Organizational Preferences. *ACM/IEEE International Conference on Human-Robot Interaction (HRI)*, Edinburgh, UK.

2026 J.S. Ong, **L. Meng**, B. Tidd, Y. Chen, D. Kulić, H. Chung. Towards Perception-Aware Shared Autonomy: A Multimodal Teleoperation Dataset for Fragile Object Manipulation. *IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. Dataset: doi.org/10.26180/31885033.

2023 B. Beck, T. Rathnayake, D. Kulić, M. Li, **L. Meng**, Z. Chen, A. Cosgun. Detecting near-miss events with cyclists using computer vision approaches. *Australasian Road Safety Conference*, Cairns, Australia.

2021 **L. Meng**, R. Gorbet, D. Kulić. Memory-based Deep Reinforcement Learning for POMDPs. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Prague.

2021 **L. Meng**, R. Gorbet, D. Kulić. The Effect of Multi-step Methods on Overestimation in Deep Reinforcement Learning. *25th International Conference on Pattern Recognition (ICPR)*, Milan.

2018 K. Kumagai, D. Lin, **L. Meng**, A. Blidaru, P. Beesley, D. Kulić, I. Mizuuchi. Towards individualized affective human-machine interaction. *IEEE International Symposium on Robot and Human Interactive Communication*.

Teaching & Supervision

2026 Co-designed, with Prof. Dana Kulić and CSIRO Human-Robot Interaction (HRI) colleagues, two sessions of Monash University's ECE4078 Intelligent Robotics unit (∼200 students) for Semester 2, 2026: a Reinforcement Learning session and an HRI session (both delivered).

2025–2026 Supervised 15 Final Year Project students across 5 projects and 3 semesters (2025 S1–2026 S1) at Monash University: multi-view preference learning, deep RL, VLA reinforcement learning via learned reward models (Robometer), VR-teleoperated dexterous grasping, humanoid locomotion safety. The multi-view preference learning project is being extended into a journal manuscript with a full user study. This extension is now continuing primarily as my own research.

2024–Present Co-supervising 2 PhD students at Monash University: Jin Siang Ong (perception-aware shared autonomy for fragile-object manipulation, oral presentation RO-MAN 2026) and Emmanuel Fashae (interpretable models of human organisational preferences, oral presentation ACM/IEEE HRI 2026). Both passed confirmation milestones in 2025.

2017–2023 Assisted in guiding 1 Master's student during PhD at University of Waterloo (RL experimentation procedures and analytical methodologies).

2016 Teaching Assistant (Object Oriented Program Analysis and Design), China University of Mining and Technology: lab leadership, discussions, assessment.

Service & Leadership

Conference reviewing	CoRL 2025; IROS 2025, 2024; ICRA 2026, 2024; ICPR 2024; HRI 2026, 2025.
Journal reviewing	ACM THRI; IEEE RA-L; Applied Soft Computing; IEEE TCDS.
2024–Present	Milestone Panel Examiner for PhD and Master's students at Monash University (Electrical and Computer Systems Engineering / Mechanical and Aerospace Engineering).
2024–Present	Research Reading Group Facilitator, Monash Robotics (reinforcement learning, preference learning, HRI).
2024–Present	Responsible for specific CSIRO HRI lab equipment, including the Franka Panda manipulation platform and GB300 workstation, among others. Authored Donesafe risk assessments for the lab's 3D printer and the GB300 workstation's UPS. Area Fire Warden.
2024	Technical Committee Member, Workshop on Human-aligned Reinforcement Learning for Autonomous Agents and Robots (ICRA 2024).
2021	Session Chair, Reinforcement Learning IV, IROS 2021.
2019	Volunteer, IEEE-RAS International Conference on Humanoid Robots (Humanoids 2019), Toronto.

Outreach & Impact

- **Public engagement:** conducted an RL-behaviour field study (Sept. 14–Oct. 3, 2018) on a Living Architecture Systems installation (Philip Beesley Studio) during *Philip Beesley: Transforming Space*, a major Royal Ontario Museum exhibition (Toronto, Jun–Oct 2018), documented in ACM THRI.
- **Non-profit partnership:** Amy's Foundation, provided my cyclist near-miss/close-pass detection code and trained models to support their community road-safety work.
- **Cross-disciplinary collaboration:** Living Architecture Systems Group: architects, artists, designers, sustained multi-institution international collaboration.
- **Lab tours & demonstrations:** CSIRO, Monash Robotics, University of Waterloo: visiting academics, industry partners, government officials, school students.
- **Open science:** open-source code, datasets, and Isaac Lab tutorials (novel robots and custom tasks) used by multiple research groups globally.

Awards & Scholarships

2023	DAAD AI-Net Fellowship: Human Centered AI.
2022	University of Waterloo David Johnston International Experience Award.
2022	University of Waterloo Doctoral Thesis Completion Award.
2021	University of Waterloo Graduate Studies Conference Assistantship.
2021	University of Waterloo S.P. Pasupalak Scholarship in Robotics and Artificial Intelligence.
2020–2021	University of Waterloo Graduate Scholarship (three installments).
2017–2022	University of Waterloo Graduate Research Studentship (GRS).
2017–2021	University of Waterloo International Doctoral Student Award (IDSA).
2017	Excellent Innovative Master's Scholarship, Graduate School of CUMT.
2014–2016	Second Prize Academic Scholarship, Graduate School of CUMT (two years).
2014	Outstanding Graduate Scholarship, CUMT.
2011–2013	First Prize Scholarship, CUMT (two years).
2010–2011	Second Prize Scholarship, CUMT.

Languages

Chinese Native

English Fluent *listening, speaking, reading, writing*

Skills

Python, PyTorch, Docker, Unity, MATLAB, C++, NVIDIA Isaac Lab/Isaac Sim, ROS, LaTeX.